
Seasonal Variation in Spatial Extremal Dependence of Wind Gusts in the Netherlands

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1 Introduction

The Netherlands is situated within a low-lying river delta, with a substantial percentage of its land located below sea level. The country’s relationship with extreme weather has historically been framed as a relationship with water: the flood of February 1953, known in Dutch as the *Watersnoodramp*, killed more than 1800 people in Zeeland and South Holland after dikes failed during a spring tide combined with a severe North Sea storm (Gerritsen, 2005). The subsequent Delta Plan and Dutch flood-protection regime implemented safety standards, with historical return periods ranging from 1250 years for major river areas to 10 000 years for the most densely populated coastal parts of Holland (Deltacommissie, 1960). These return periods illustrate how extreme-value reasoning has long been embedded in Dutch climate-risk management. Wind has received comparatively less attention in this Dutch climate-risk narrative, despite the fact that 10 out of 20 most costly climate related hazards were windstorms (Verbond van Verzekeraars, 2017). Investigating seasonal differences in spatial extremal dependence is relevant because the societal impact of wind extremes depends not only on the magnitude of gusts at individual locations, but also on the spatial extent of the event.

Existing windstorm studies often focus on marginal extremes, return levels, or losses. For national risk, however, the relevant question is not only how large the gust is at one station, but how many other stations are extreme at the same time. Our paper addresses this gap by focusing on pairwise extremal dependence rather than only marginal wind-gust behaviour.

Wind extremes are spatial events. A severe winter storm associated with a mature extratropical cyclone may produce damaging gusts across large parts of the Netherlands within a few hours, reflecting the large spatial footprints commonly associated with European windstorms (Sharkey et al., 2020; Little et al., 2023). By contrast, summer convective gusts are often linked to thunderstorms and local-scale downbursts, whose small-scale and non-stationary nature can generate sharp spatial differences in observed gust intensity (Mohr et al., 2017). These two events may look similar from the perspective of a single-station maximum, but they imply very different risks for society. The first is a system-wide event, with simultaneous pressure on infrastructure, insurers, transport networks and emergency services; the second is more localised, but potentially harder to anticipate at fine spatial scales (Gliksman et al., 2023; Fonseca Cerda et al., 2025). This distinction motivates the focus on spatial extremal dependence rather than only marginal wind-gust extremes.

Spatial extreme value theory provides a general framework for analysing such tail dependencies. The pairwise extremal coefficient θ_{ij} and the upper-tail dependence coefficient χ_{ij} summarise the joint behaviour of a station pair at the upper tail (Davison et al., 2012; Cooley, Cisewski, et al., 2012). The extremal coefficient θ_{ij} lies between one and two, with values close to one corresponding to strong extremal dependence and values close to two corresponding to weak extremal dependence. The upper-tail dependence coefficient χ_{ij} describes the limiting probability that one station is extreme conditional on another station being extreme; because this limit is not observable in finite data, our paper works with the finite-level coefficient χ_u , evaluated at high thresholds. Both coefficients are functions of the distance between stations, so the spatial extremal dependence structure can be summarised as a dependence–distance curve.

The methodological problem is therefore not only to describe wind-gust extremes, but to determine whether seasonal differences in spatial tail dependence can be estimated reliably from a finite and spatially irregular station network. Splitting the sample by season reduces the effective number of extreme observations, while estimating dependence for many station pairs introduces additional sampling variability across distances. We therefore frame the empirical comparison between winter and summer as a problem of finite-sample inference for spatial extremal dependence. Because the true dependence structure of the observations is unknown, a simulation study based on a Brown–Resnick max-stable benchmark process, whose dependence–distance relation is known by construction, is used to examine whether the selected pairwise estimators can recover known seasonal differences in extremal dependence before the same estimators are applied to the KNMI observations.

1.1 Research question

The main research question of our paper is:

Is there more spatial extremal dependence of wind gusts between KNMI station locations in winter than in summer in the Netherlands over the period 1991–2025?

The comparison is between winter, defined as December–February (DJF), and summer, defined as June–August (JJA). The hypothesis is that winter wind-gust extremes are more spatially coherent than

summer wind-gust extremes. In terms of the extremal coefficient, this corresponds to lower values of θ_{ij} for winter than for summer at comparable inter-station distances. In terms of the finite-level upper-tail dependence coefficient, it corresponds to higher values of χ_u . The empirical analysis therefore compares season-specific dependence–distance curves rather than only testing for a difference at a single station pair.

1.2 Data and empirical workflow

The empirical application uses hourly observations from the network of automatic weather stations operated by the Royal Netherlands Meteorological Institute (KNMI). The hourly maximum wind gust, denoted FX, is expressed in metres per second. Stations are indexed by their location $s_i \in \mathbb{R}^2$, with latitude, longitude and a sensor-height attribute recorded in the station metadata. The full dataset covers the years from 1951 to 2026. Due to station-network expansion and missing observations, the number of stations reporting FX in any given hour varies with time. The analysis sample is 1991–2025, which is the first 35-year period with a sufficiently large and stable station network of around 25 stations.

The empirical workflow proceeds in steps. The hourly observations are first aggregated to daily maximum wind gusts at each station, which reduces the most immediate within-day serial dependence while retaining far more observations than seasonal maxima. The daily maxima are split into winter (DJF) and summer (JJA), and the margins are transformed separately by station and season to a common empirical uniform scale, so that the winter–summer comparison reflects differences in spatial dependence rather than differences in the marginal distributions. The pairwise F -madogram, extremal coefficient θ_{ij} , and finite-level χ_u are then estimated for each station pair and summarised as season-specific dependence–distance curves. Uncertainty in these curves is assessed by a year or seasonal-block bootstrap, which accounts for the residual serial dependence between consecutive daily maxima within a storm.

1.3 Contribution

The contribution of our paper is to study the finite-sample performance of pairwise estimators of spatial extremal dependence in a seasonal comparison design. Methodologically, a simulation study based on a Brown–Resnick benchmark process establishes whether these estimators can recover a known winter-versus-summer difference in spatial extremal dependence from finite daily maxima observed on a fixed, irregular station network. Empirically, the same estimators are then applied to the KNMI wind-gust data, comparing two complementary dependence summaries, the extremal coefficient θ and the finite-level tail-dependence coefficient χ_u , across inter-station distances in order to assess whether winter and summer dependence–distance curves differ.

2 Literature Review

This section reviews the literature on extreme-value theory for environmental data, spatial extremes and max-stable processes, pairwise extremal-dependence measures, data-generating processes for the simulation study, inference and uncertainty for dependence estimators, and seasonal and non-stationary extremal dependence.

2.1 Extreme-value theory

Extreme-value theory provides the statistical framework for studying rare events in the tails of a distribution. For the canonical case of maxima, let $X_1, \dots, X_n$ be independent observations with common distribution function F and let $M_n = \max\{X_1, \dots, X_n\}$. Then

$$\Pr\{M_n \leq z\} = \Pr\{X_1 \leq z, \dots, X_n \leq z\} = F(z)^n.$$

The results of Fisher and Tippett (1928) and Gnedenko (1943) show that, under normalisation, any non-degenerate limit of normalised sample maxima must belong to the generalised extreme value (GEV) family, of which the Gumbel, Fréchet, and Weibull types are special cases. This justifies modelling tail observations directly and provides the basis for the block-maxima approach, in which the data are divided into non-overlapping blocks and the maximum within each block is modelled. Block maxima are also the entry point for spatial extremes, since spatial max-stable processes arise as the limiting objects for spatial block maxima (Laurens de Haan and Pereira, 2006).

In our paper the blocks are days: the hourly observations are aggregated to daily maximum wind gusts at each station, and these are the data object for all subsequent dependence estimation. Daily maxima sit between two extremes of block length. Raw hourly gusts are strongly serially dependent, as consecutive hours typically belong to the same storm system, whereas seasonal maxima would leave only about 35 observations per station per season over 1991–2025, too few to estimate pairwise dependence stably. Daily maxima retain on the order of ninety observations per station per season while reducing the most immediate within-day clustering.

An alternative is the peaks-over-threshold (POT) framework, in which exceedances above a high threshold are approximated by a generalised Pareto distribution (Balkema and L. de Haan, 1974; Pickands, 1975). For hourly data, consecutive exceedances usually belong to the same storm and require declustering (Ferro and Segers, 2003) or explicit temporal modelling (Fawcett and Walshaw, 2006). Because daily aggregation already addresses the most immediate serial dependence, our paper treats POT and declustering as robustness alternatives rather than the main strategy.

The marginal step is only the first part of the problem. The spatial question is: conditional on one station observing an extreme daily maximum wind gust, how likely is another station to observe an extreme value on the same day? This shifts the focus to spatial extremal dependence, whose estimands are pairwise tail-dependence measures, extremal coefficients, and dependence–distance curves.

2.2 Spatial extremes and max-stable processes

For block maxima observed over space, max-stable processes provide the standard limiting framework, arising as the limit of pointwise maxima of independent replications of a spatial random field, just as the GEV distribution arises as the limit of scalar maxima. Extremal dependence thereby becomes a spatial object, described as a function of location and distance. Laurens de Haan and Pereira (2006) give a foundational treatment of stationary max-stable models, showing how spatial extremal dependence can be governed by a few parameters related to inter-station distance. Their stationarity assumption imposes a homogeneous dependence structure, which may be restrictive for Dutch gusts, since winter and summer storms need not share the same spatial pattern.

Davison et al. (2012) review the main approaches to spatial extremes and conclude that good marginal modelling is not sufficient: models that fit univariate extremes well may still misrepresent joint extremes across locations. This motivates our focus on extremal-dependence summaries rather than marginal behaviour alone. Full-likelihood inference for max-stable processes is infeasible once the number of locations is moderate, and Padoan et al. (2010) replace it with a pairwise composite likelihood, establishing pairwise inference as a standard approach.

In our paper, max-stable processes serve mainly as a theoretical reference point and as the benchmark data-generating process for the simulation study (Section 2.4), not as a model fitted to the data. We adopt the pairwise logic of Padoan et al. (2010), but the empirical objects are nonparametric pairwise summaries computed directly from the seasonal daily maximum wind gusts.

2.3 Pairwise extremal-dependence measures

The analysis requires dependence measures interpretable at the station-pair level. Cooley, Cisevski, et al. (2012) survey such tools, including extremal coefficients, madograms, and tail-dependence summaries, which reduce the high-dimensional dependence problem to pairwise quantities that can be compared across seasons and plotted against distance. In our paper all three are estimated from the seasonal daily maximum wind gusts, separately for winter (DJF) and summer (JJA), with margins transformed separately by station and season so that the comparison reflects differences in spatial dependence rather than in the marginal distributions.

For two locations s_i and s_j , the pairwise extremal coefficient $\theta(s_i, s_j)$ lies in $[1, 2]$: values close to one indicate strong extremal dependence and values close to two near-independence, so stronger winter dependence corresponds to lower winter $\theta(s_i, s_j)$ at comparable distances. An empirical route to it is the F -madogram. Cooley, Naveau, et al. (2006) study madogram-based summaries for max-stable random fields; for transformed margins,

$$\nu_F(s_i, s_j) = \frac{1}{2} \mathbb{E} \left[\left| F_{s_i} \{X(s_i)\} - F_{s_j} \{X(s_j)\} \right| \right],$$

where F_{s_i} and F_{s_j} are the marginal distribution functions, and under max-stability

$$\theta(s_i, s_j) = \frac{1 + 2 \nu_F(s_i, s_j)}{1 - 2 \nu_F(s_i, s_j)}.$$

The F -madogram thus gives a direct nonparametric estimate of a dependence measure summarised as a dependence–distance curve. The upper-tail dependence coefficient χ gives a complementary view. With $U(s_i)$ and $U(s_j)$ the marginally standardised observations on a uniform scale,

$$\chi(s_i, s_j) = \lim_{u \uparrow 1} \Pr\{U(s_j) > u \mid U(s_i) > u\},$$

and the finite-level quantity is

$$\chi_u(s_i, s_j) = \Pr\{U(s_j) > u \mid U(s_i) > u\}.$$

Computed within each season from daily co-exceedances, χ_u answers the research question directly: if one KNMI station records an extreme daily maximum wind gust, how likely is another to do so on the same day? Larger χ_u indicates stronger finite-level co-exceedance, just as smaller θ indicates stronger extremal dependence. Coles et al. (1999) introduce χ and $\bar{\chi}$ to distinguish asymptotic dependence from asymptotic independence and discuss their finite-level estimation. Wadsworth and Tawn (2012) and Huser and Wadsworth (2022) caution that classical max-stable models are natural for asymptotic dependence but that finite data may show weakening dependence at increasingly high levels; our paper therefore reports finite-level dependence on a small grid of thresholds rather than a single threshold.

2.4 Data-generating processes for the simulation study

To check whether the estimators recover a true dependence structure, recovery must be measured against a known dependence–distance relation. The pairwise extremal coefficients of max-stable processes are available in closed form, so a simulated field carries a known $\theta(s_i, s_j)$ at every distance against which the estimated curve can be compared. The Brown–Resnick process (Brown and Resnick, 1977), characterised through a negative-definite variogram, is a widely used family whose extremal dependence is controlled by that variogram and ranges from strong to near-independent as its range varies, and Dombry et al. (2016) provide exact simulation algorithms that ensure the known dependence structure is the one actually simulated.

Our paper uses the Brown–Resnick process as the benchmark data-generating process, since its variogram gives direct control over how extremal dependence decays with distance. The null and alternative designs set its variogram-range parameters so that the two regimes either share a dependence structure or differ by a controlled amount. The Brown–Resnick process is used only as a controlled benchmark whose true dependence is known; we do not assume that the Dutch wind-gust data are Brown–Resnick. The aim is solely to test whether the pairwise estimators recover a known winter-versus-summer difference in spatial extremal dependence.

2.5 Inference and uncertainty for dependence estimators

Because our paper compares estimated dependence–distance curves, a difference between two curves is only informative once the uncertainty in each is quantified. Two sources of variability matter. First, the F -madogram and χ_u estimators are empirical means and conditional frequencies computed from a finite number of seasonal daily maxima, so their variance grows as the season-split sample shrinks. Second, although daily aggregation removes within-day clustering, a multi-day storm can produce extreme daily maxima on consecutive days, so daily observations cannot be treated as fully independent.

Our paper therefore quantifies uncertainty by resampling temporal blocks. A year or seasonal-block bootstrap resamples whole years, or whole seasonal segments, with replacement and recomputes the full dependence–distance curves on each resample, so that the resulting variability reflects both the finite sample size and the residual serial dependence within storms. The composite-likelihood framework of Padoan et al. (2010) provides asymptotic standard errors but relies on correct parametric specification; the nonparametric summaries used here are instead assessed by block resampling and, in the simulation, by Monte Carlo evaluation against the known truth.

2.6 Seasonal and non-stationary extremal dependence

The seasonal question is a restricted form of non-stationary spatial extremal dependence: instead of one dependence structure for all daily maxima, our paper lets the dependence relation differ between two regimes, with season treated as a binary covariate. Huser and Genton (2016) develop non-stationary dependence structures for spatial extremes, showing that extremal dependence itself, not only the marginal

distribution, can vary with covariates. The seasonal split is a deliberately simple version of this idea and also addresses a stationarity concern: estimating one dependence curve from the unstratified daily maxima would mix large-scale winter storms with more localised summer convective events. Stratifying by season does not remove all non-stationarity, but it makes the working assumption more credible and gives each estimated curve a clearer interpretation.

In summary, Fisher and Tippet (1928), Gnedenko (1943), Balkema and L. de Haan (1974), and Pickands (1975) justify modelling tail observations and motivate the daily block-maxima choice; Laurens de Haan and Pereira (2006), Davison et al. (2012), and Padoan et al. (2010) provide the max-stable and pairwise-inference framework; Cooley, Naveau, et al. (2006), Cooley, Cisewski, et al. (2012), and Coles et al. (1999) motivate the pairwise estimands, with Wadsworth and Tawn (2012) and Huser and Wadsworth (2022) motivating the threshold-grid robustness check; the Brown–Resnick process (Brown and Resnick, 1977) with exact simulation (Dombry et al., 2016) supplies the benchmark for validation; and Huser and Genton (2016) frames the seasonal comparison as non-stationary extremal dependence. The contribution of our paper is to combine these in a deliberately narrow setting: to establish, by simulation, when a seasonal difference in spatial extremal dependence can be recovered from finite daily maximum wind gusts on a fixed, irregular station network, and then to apply the validated estimators to Dutch wind gusts.

3 Methodology

This section sets out the empirical strategy used to answer the seasonal extremal-dependence question of Section 1.1. The strategy has five ingredients: (i) construction of daily maximum wind gusts from the hourly FX series; (ii) seasonal stratification into winter and summer; (iii) marginal standardisation, separately by station and season; (iv) estimation of pairwise extremal-dependence summaries; and (v) summarisation as dependence–distance curves, with uncertainty from a year or seasonal-block bootstrap. Operational numerical choices, such as the threshold quantile, the bootstrap block length, and the smoother bandwidth, are recorded in Section ??, so that the methodology remains valid when those values are tuned.

3.1 Overview of the empirical strategy

The goal is to estimate, separately for winter and summer, the dependence–distance curves in the extremal coefficient and in the finite-level tail-dependence coefficient,

$$\theta^{(c)}(d), \quad \chi_u^{(c)}(d), \quad c \in \{W, S\},$$

where d is the inter-station distance and c indexes the season. The seasonal comparison is then the pair of estimated curves displayed on common axes with bootstrap uncertainty bands, supplemented by a pointwise comparison of the winter and summer curves.

The data object is the *daily maximum wind gust*. The hourly FX series is aggregated to one maximum per station per day, so that each daily value is a one-day block maximum in the sense of the block-maxima theory of Section 2.1. This choice is the methodological core of the design: daily aggregation removes the most immediate within-day serial dependence of hourly gusts, which belong to the same passing storm system, while retaining far more observations than seasonal maxima would. As discussed in Section 2.1, seasonal maxima would leave only about 35 observations per station per season over 1991–2025, too few to estimate pairwise dependence stably, whereas daily maxima retain on the order of ninety observations per station per season. Peaks-over-threshold construction with declustering is retained only as a robustness alternative (Section 3.5), consistent with the treatment in Section 2.1.

3.2 Station and sample selection

We use the fixed panel of around 25 KNMI land and coastal stations that report FX over the analysis window 1991–2025. The following filters are applied in order.

1. Keep only Dutch land and coastal KNMI stations reporting FX (Section ??).
2. Discard hours where FX is missing or flagged invalid by KNMI quality control, before daily aggregation.
3. Discard a station-day if its within-day missing fraction exceeds the completeness threshold of Section ??, so that each retained daily maximum is based on adequate hourly coverage.
4. Discard a station-season-year that fails the within-season completeness threshold of Section ??.

Missing values exist but are not a central methodological issue on this fixed network; pairwise estimates are computed on a pairwise-complete basis, so a pair contributes on every day for which both stations report a valid daily maximum.

3.3 Seasonal stratification

We split the daily series into two non-overlapping seasonal subsamples,

$$\mathcal{T}_W = \{t : \text{month}(t) \in \{12, 1, 2\}\}, \quad \mathcal{T}_S = \{t : \text{month}(t) \in \{6, 7, 8\}\},$$

where t now indexes days. December of calendar year y is attached to winter year $y + 1$, so that each winter is contiguous in calendar time. The transitional seasons (MAM and SON) are dropped from the main analysis. As argued in Section 2.6, this stratification treats season as a binary covariate for the dependence structure and makes the within-season working stationarity assumption more credible than pooling all daily maxima, which would mix large-scale winter storms with more localised summer convection.

3.4 Construction of daily maximum wind gusts (main route)

For each station i and each day t , the daily maximum wind gust is

$$X_{i,t} = \max\{FX_{i,h} : h \in \text{day}(t)\},$$

where the maximum is taken over the (up to 24) valid hourly observations within day t . The series $\{X_{i,t} : t \in \mathcal{T}_c\}$ of seasonal daily maxima is the input to all dependence estimation below. Each $X_{i,t}$ is a one-day block maximum, so the block-maxima interpretation of Section 2.1 applies directly with the block fixed at one calendar day.

3.5 Construction of extreme observations (robustness route: threshold exceedances)

As a robustness alternative we also construct, for each station i and season c , peaks over a high station- and season-specific threshold $u_i^{(c)}$, set as a fixed empirical quantile of $\{X_{i,t} : t \in \mathcal{T}_c\}$. Exceedances are

$$Y_{i,t}^{(c)} = X_{i,t} - u_i^{(c)} \quad \text{for } X_{i,t} > u_i^{(c)}, \quad t \in \mathcal{T}_c.$$

Because a single multi-day storm can produce exceedances on consecutive days, the exceedance sequence is declustered using the automatic interexceedance-time scheme of Ferro and Segers (2003), yielding an approximately independent set of cluster peaks per station and season; Fawcett and Walshaw (2006) provide an alternative treatment that models the short-range temporal dependence of exceedances directly. This route is reported only as a robustness check (Section 3.11); daily aggregation already addresses the most immediate within-day serial dependence, so the primary analysis works from daily maxima rather than declustered hourly exceedances, in line with Section 2.1.

3.6 Marginal standardisation

To ensure that the winter–summer comparison reflects differences in spatial dependence rather than in the marginal distributions, we transform each station-season series to a common $(0, 1)$ margin. Coastal and inland stations have different local gust climates, and winter and summer differ in their marginal gust distributions, so the transform is applied separately by station *and* season.

Empirical rank transform (main). For each station i and season c , set

$$\widehat{U}_{i,t}^{(c)} = \widehat{F}_i^{(c)}(X_{i,t}), \quad t \in \mathcal{T}_c,$$

where $\widehat{F}_i^{(c)}$ is the within-season empirical distribution function of the daily maxima at station i . This is the default because it makes no parametric assumption about the marginal distribution.

Parametric tail transform (robustness). For each station and season, fit a GEV to the seasonal daily maxima $\{X_{i,t} : t \in \mathcal{T}_c\}$ and transform to a unit-Fréchet margin via $Z_{i,t}^{(c)} = -1/\log \widehat{G}_i^{(c)}(X_{i,t})$. This exploits the extreme-value limit but is sensitive to the marginal fit, and is reported as a robustness check (Section 3.11).

3.7 Pairwise extremal-dependence estimation

We estimate the season-stratified pairwise summaries introduced in Section 2.3: the extremal coefficient $\theta_{ij}^{(c)}$ via the F -madogram, the finite-level tail-dependence coefficient $\chi_{u,ij}^{(c)}$, and the companion diagnostic $\bar{\chi}_{ij}^{(c)}$. All estimators are empirical and apply directly to the rank-transformed seasonal daily maxima $\widehat{U}_{i,t}^{(c)}$.

Extremal coefficient via the F -madogram. Following Cooley, Naveau, et al. (2006), for each pair (i, j) and season c the F -madogram is estimated on the seasonal daily maxima by

$$\widehat{\nu}_F^{(c)}(s_i, s_j) = \frac{1}{2n_{ij}^{(c)}} \sum_{t \in \mathcal{T}_c^{ij}} |\widehat{U}_{i,t}^{(c)} - \widehat{U}_{j,t}^{(c)}|,$$

where $\mathcal{T}_c^{ij}$ is the set of days on which both stations report a valid daily maximum and $n_{ij}^{(c)} = |\mathcal{T}_c^{ij}|$. The extremal coefficient is recovered by the max-stable relation of Section 2.3,

$$\hat{\theta}_{ij}^{(c)} = \frac{1 + 2\hat{\nu}_F^{(c)}(s_i, s_j)}{1 - 2\hat{\nu}_F^{(c)}(s_i, s_j)}.$$

Finite-level tail-dependence coefficient. For each pair (i, j) , season c , and threshold u on a grid $\mathcal{U}$ close to one, the conditional co-exceedance probability is estimated by

$$\hat{\chi}_{u,ij}^{(c)} = \frac{\#\{t \in \mathcal{T}_c^{ij} : \hat{U}_{i,t}^{(c)} > u, \hat{U}_{j,t}^{(c)} > u\}}{\#\{t \in \mathcal{T}_c^{ij} : \hat{U}_{i,t}^{(c)} > u\}}.$$

Because this ratio is directional in finite samples, the implementation uses the symmetrised version that averages the two conditioning directions. Following Wadsworth and Tawn (2012) and Huser and Wadsworth (2022), $\hat{\chi}_{u,ij}^{(c)}$ is reported on a grid of u rather than at a single level, so that weakening dependence at the very top of the tail can be detected; the reference level used in the headline comparison is set in Section ??.

Asymptotic-independence diagnostic. For each pair we also compute the finite-level $\hat{\chi}_{ij}^{(c)}(u)$ on the same grid, reported jointly with $\hat{\chi}_{u,ij}^{(c)}$ to diagnose whether a pair is better described as asymptotically dependent or asymptotically independent (Coles et al., 1999).

3.8 Distance summary

The pair-level estimates are noisy, so for each season we summarise them against the inter-station distance $d_{ij} = \text{dist}(s_i, s_j)$. Writing $\psi_{ij}^{(c)}$ for either $\hat{\theta}_{ij}^{(c)}$ or $\hat{\chi}_{u,ij}^{(c)}$, the dependence–distance curve is

$$\hat{\psi}^{(c)}(d) = \text{smoother}\left(\{(d_{ij}, \psi_{ij}^{(c)})\}_{i < j}\right),$$

estimated by a local-linear kernel smoother with Gaussian kernel and a bandwidth selected within each season; the bandwidth rule and value are reported in Section ??. The smoother is a device to make the seasonal comparison legible, not an object of interest in itself, so the raw pairwise scatter is always shown alongside the smoothed curve so that the reader can judge whether the smoother hides structure.

3.9 Winter–summer comparison

The headline comparison combines three reports, computed for both θ and χ_u .

1. **Overlay.** The winter and summer smoothed curves $\hat{\psi}^{(W)}(d)$ and $\hat{\psi}^{(S)}(d)$ on a single panel with their bootstrap bands.
2. **Scalar summaries.** The curve values at a small set of reference distances d_0 (for example 50, 100, and 200 km), each with a bootstrap percentile interval.
3. **Difference curve.** The pointwise difference $\hat{\psi}^{(W)}(d) - \hat{\psi}^{(S)}(d)$ with a bootstrap band. Under the hypothesis of stronger winter dependence, this difference is negative for θ and positive for χ_u at comparable distances.

The comparison is read off the bootstrap bands of the difference curve rather than from a single hypothesis test, consistent with the uncertainty treatment in Section 2.5.

3.10 Uncertainty quantification

As argued in Section 2.5, two sources of variability matter: the finite number of seasonal daily maxima, which inflates the variance of the empirical estimators as the season-split sample shrinks; and the residual serial dependence between consecutive daily maxima, since a multi-day storm can produce extreme daily maxima on several consecutive days. Daily aggregation removes within-day clustering but not this multi-day dependence, so the daily observations cannot be treated as fully independent.

We therefore quantify uncertainty by a *year (or seasonal-block) bootstrap*. Whole years, or whole seasonal segments, are resampled with replacement; within each bootstrap replication the marginal rank transform and all pairwise estimators are recomputed, and the dependence–distance curves are re-estimated. Percentile intervals are formed over B replications. Resampling whole temporal blocks, rather than individual days, preserves the within-storm serial dependence, so the resulting bands reflect both the finite sample size and that residual dependence. The number of replications B is recorded in Section ???. The pairwise composite-likelihood standard errors of Padoan et al. (2010) are available for the optional parametric fit below, but they rely on correct parametric specification and are not used for the nonparametric summaries.

3.11 Robustness checks

The robustness panel (Section ??) comprises:

1. Threshold-exceedance route with declustering as an alternative to daily block maxima.
2. Marginal standardisation: empirical rank transform vs. parametric GEV tail transform.
3. Threshold level for χ_u : sensitivity across the grid $\mathcal{U}$.
4. Calendar-time sample split (early vs. late half) as a secondary long-run analysis.
5. Anisotropy: allowing the dependence to depend on direction as well as scalar distance.

3.12 Optional: parametric max-stable fitting

If time permits, we fit a parametric max-stable process, for example the Brown–Resnick family (Brown and Resnick, 1977), to the seasonal daily maxima by pairwise composite likelihood (Padoan et al., 2010), one season at a time, and compare the fitted dependence function with the nonparametric $\hat{\theta}^{(c)}(d)$ of Section 3.8. As stressed in Section 2.2, the max-stable process is used here only as a parametric reference point, not as a maintained assumption about the data. A fully covariate-dependent, season-indexed max-stable model in the spirit of Huser and Genton (2016) is beyond the scope of our paper and is recorded only as a future-work direction in Section ??.

3.13 Limitations

Three limitations are worth flagging.

- *Within-season stationarity.* The strategy treats the field as approximately stationary within each season. This is the reason for the seasonal stratification, but it is not exact: residual non-stationarity within a single DJF season remains, which the calendar-time robustness split addresses only partially.
- *Asymptotic dependence.* The estimators treat finite-level quantities as informative about the underlying extremal dependence. As Wadsworth and Tawn (2012) and Huser and Wadsworth (2022) note, finite-level dependence can weaken at the very top of the tail; we address this by reporting χ_u on a grid of u and by including the $\bar{\chi}$ diagnostic.
- *Isotropy.* The distance smoother treats dependence as a function of scalar distance only. Anisotropy due to prevailing storm direction is a plausible second-order effect, examined as a robustness check in Section 3.11.

4 Simulation Study

The empirical analysis of this paper compares winter and summer spatial extremal dependence using pairwise nonparametric estimators. Before applying these estimators to the KNMI daily maximum wind-gust data, it is useful to study their finite-sample behaviour in a controlled setting where the true dependence structure is known. The purpose of the simulation study is therefore methodological: it assesses whether the proposed estimators are able to recover differences in spatial extremal dependence of the type expected in the empirical application. In particular, the simulation investigates whether the estimators can detect a regime in which winter extremes are more strongly and more widely spatially dependent than summer extremes.

The simulation is not intended to show that Dutch wind-gust extremes are generated by a specific parametric max-stable model. Instead, a max-stable benchmark is used as a controlled data-generating process. This allows the true pairwise extremal coefficient to be computed for every station pair and compared with the estimated dependence measures. The simulation therefore acts as a validation step for the empirical methodology: if the estimators fail to recover known dependence differences in finite samples comparable to the KNMI data, their interpretation in the empirical section would be questionable.

4.1 Station geometry and sample size

The simulated spatial fields are generated on the same type of station geometry as the empirical analysis. Let $(s_1, \dots, s_d)$ denote the retained KNMI station locations, where d is the number of stations in the fixed panel. For each station pair (i, j) , the inter-station distance is denoted by

$$h_{ij} = |s_i - s_j|,$$

measured in kilometres. The use of the empirical station geometry ensures that the simulation reflects the same range of inter-station distances as the real-data analysis.

The simulated sample size is chosen to mimic the daily-maxima design. For each regime, a sample of n spatial extreme fields is generated, where n corresponds approximately to the number of seasonal daily maxima available in the empirical sample, pooled across all years. With 35 years of data and roughly 90 days per season, each station-season contains approximately

$$n \approx 35 \times 90 = 3150$$

daily maxima before accounting for missing values. This is the total pooled over all winters (or all summers), not the count within a single year, since the empirical estimators pool seasonal daily maxima across the whole window. The simulation therefore uses this order of sample size as its baseline setting, while smaller values may be used in preliminary runs to assess computation time and to study how estimator performance degrades as the effective tail sample shrinks.

4.2 Brown–Resnick benchmark process

The main benchmark data-generating process is the Brown–Resnick max-stable process. This process is well suited to the present simulation because its extremal dependence is controlled by a variogram, which gives a direct way to vary the spatial range of extremal dependence. For two locations separated by distance h , the semivariogram is specified as

$$\gamma(h) = \left(\frac{h}{\rho}\right)^\alpha,$$

where $\rho > 0$ is a range parameter and $0 < \alpha \leq 2$ controls the smoothness of the dependence decay. Larger values of ρ imply slower growth of the variogram with distance and hence stronger dependence at a given distance.

For the Brown–Resnick process, the pairwise extremal coefficient is

$$\theta(h) = 2\Phi\left(\frac{\sqrt{\gamma(h)}}{2}\right),$$

where Φ denotes the standard normal distribution function. This expression gives the true dependence-distance curve against which the estimators can be evaluated. Since $\theta(h) \in [1, 2]$, values close to one indicate strong extremal dependence, while values close to two indicate near extremal independence.

The simulation considers two regimes, labelled winter and summer. These labels are used only to mirror the empirical comparison. In the null design, both regimes have the same variogram parameters:

$$\rho_{\text{winter}} = \rho_{\text{summer}}.$$

Therefore, there is no true seasonal difference in spatial extremal dependence. Under the alternative design, winter is assigned a larger range parameter:

$$\rho_{\text{winter}} > \rho_{\text{summer}}.$$

This implies

$$\theta_{\text{winter}}(h) < \theta_{\text{summer}}(h)$$

for relevant distances $h > 0$, meaning that winter extremes are more spatially dependent and that this dependence persists over longer distances.

A baseline configuration uses $\alpha = 1$. Under the null, both regimes may use $\rho = 120$ km. Under the alternative, a representative setting is

$$\rho_{\text{winter}} = 180 \text{ km}, \quad \rho_{\text{summer}} = 60 \text{ km}.$$

These values are not treated as empirical estimates. They are chosen to create a controlled and interpretable difference in dependence range.

4.3 Estimators evaluated

The simulation evaluates the same pairwise estimators used in the empirical analysis. For each Monte Carlo replication and each regime, the simulated observations are first transformed to empirical uniform margins separately by station. This matches the empirical workflow, where station-specific marginal distributions are removed before estimating spatial dependence.

Let $U_t(s_i)$ denote the transformed observation at station s_i on simulated day t . For each pair (i, j) , the empirical F -madogram is estimated by

$$\hat{\nu}_F(s_i, s_j) = \frac{1}{2n} \sum_{t=1}^n |U_t(s_i) - U_t(s_j)|.$$

The corresponding madogram-based extremal coefficient estimator is

$$\hat{\theta}(s_i, s_j) = \frac{1 + 2\hat{\nu}_F(s_i, s_j)}{1 - 2\hat{\nu}_F(s_i, s_j)}.$$

The finite-level tail-dependence coefficient is estimated at thresholds $u \in \mathcal{U}$, for example $\mathcal{U} = \{0.90, 0.95, 0.98\}$. For a given threshold u , the conditional co-exceedance probability is

$$\hat{\chi}_u(s_i, s_j) = \frac{\sum_{t=1}^n \mathbf{1}\{U_t(s_i) > u, U_t(s_j) > u\}}{\sum_{t=1}^n \mathbf{1}\{U_t(s_i) > u\}}.$$

Because this quantity is directional in finite samples, the implementation may use a symmetrised version,

$$\hat{\chi}_u^{\text{sym}}(s_i, s_j) = \frac{1}{2} \left[\widehat{\Pr}\{U(s_j) > u \mid U(s_i) > u\} + \widehat{\Pr}\{U(s_i) > u \mid U(s_j) > u\} \right].$$

Larger values of $\hat{\chi}_u$ indicate stronger finite-level co-exceedance, while smaller values of $\hat{\theta}$ indicate stronger extremal dependence.

4.4 Monte Carlo design

Each Monte Carlo replication proceeds as follows. First, one winter sample and one summer sample are simulated on the fixed station set as independent draws of the max-stable field. This is an idealisation relative to the empirical data, where consecutive daily maxima within a multi-day storm are serially dependent; the simulation deliberately isolates the spatial estimation problem by treating the daily fields as independent replications, so that estimator bias and variability can be studied against a known truth without the additional complication of temporal dependence. The consequences of that residual temporal

dependence are handled separately, in the empirical analysis, by the year or seasonal-block bootstrap of Section 3.10; an extension that simulates temporally dependent daily maxima is noted in Section 4.6. Second, each station margin is transformed to the empirical uniform scale within each regime. Third, pairwise estimates of θ and χ_u are computed for all station pairs. Fourth, the estimated winter–summer differences are computed:

$$\Delta_{\theta,ij} = \hat{\theta}_{\text{winter}}(s_i, s_j) - \hat{\theta}_{\text{summer}}(s_i, s_j),$$

and

$$\Delta_{\chi_u,ij} = \hat{\chi}_{u,\text{winter}}(s_i, s_j) - \hat{\chi}_{u,\text{summer}}(s_i, s_j).$$

Under the alternative design, the expected signs are

$$\Delta_{\theta,ij} < 0 \quad \text{and} \quad \Delta_{\chi_u,ij} > 0,$$

because winter is designed to have stronger extremal dependence. Under the null design, both differences should be centred around zero.

The pairwise estimates are also summarised over distance bins. Let $(B_1, \dots, B_K)$ denote distance intervals, such as 0–50 km, 50–100 km, 100–150 km, 150–200 km, and above 200 km. For each bin, the simulation reports the average estimated dependence, the average theoretical dependence, and the winter–summer difference. This converts the pairwise output into dependence-distance curves, matching the empirical analysis.

4.5 Performance measures

The simulation evaluates estimator performance along several dimensions. First, for the extremal coefficient, the bias for a station pair is computed as

$$\text{Bias}_{ij} = \mathbb{E}_{\text{MC}} \left[\hat{\theta}(s_i, s_j) \right] - \theta(h_{ij}),$$

where $\mathbb{E}_{\text{MC}}$ denotes the Monte Carlo average. The corresponding root mean squared error is

$$\text{RMSE}_{ij} = \left\{ \mathbb{E}_{\text{MC}} \left[\left(\hat{\theta}(s_i, s_j) - \theta(h_{ij}) \right)^2 \right] \right\}^{1/2}.$$

These quantities show whether the madogram-based extremal coefficient estimator accurately recovers the true Brown–Resnick dependence curve.

Second, the simulation records whether the estimators recover the correct seasonal ordering. Under the alternative, the winter extremal coefficient should be lower than the summer extremal coefficient. Therefore, one diagnostic is the proportion of station pairs for which

$$\hat{\theta}_{\text{winter}}(s_i, s_j) < \hat{\theta}_{\text{summer}}(s_i, s_j).$$

Analogously, for χ_u , the correct ordering under the alternative is

$$\hat{\chi}_{u,\text{winter}}(s_i, s_j) > \hat{\chi}_{u,\text{summer}}(s_i, s_j).$$

These diagnostics directly answer the methodological question of whether the empirical estimators are able to detect stronger winter dependence when it is present by construction.

Third, the null design is used to assess false seasonal differences. Since the two regimes have the same dependence structure under the null, the average estimated winter–summer difference should be close to zero. A systematic nonzero difference under the null would indicate that the procedure can create artificial seasonal differences even when none exist.

Fourth, the simulation checks the calibration of the uncertainty bands used in the empirical analysis. For a subset of replications, the year or seasonal-block bootstrap of Section 3.10 is applied to the simulated sample, and the empirical coverage of the nominal bootstrap interval is recorded, that is the proportion of replications in which the interval for $\theta(h_{ij})$ or for the winter–summer difference contains the known true value. Coverage close to the nominal level indicates that the bootstrap bands reported in the empirical section are trustworthy at the available sample size and station geometry; coverage well below nominal would warn that the empirical bands are too narrow. Because the simulated daily fields are independent, this exercise checks the bootstrap under the favourable case of no residual temporal dependence; the temporally dependent case is left as the extension noted in Section 4.6.

4.6 Interpretation for the empirical analysis

The simulation study is used to guide the empirical interpretation in two ways. First, it indicates whether the proposed estimators are reliable at the sample size and station geometry available in the KNMI data. If the estimators recover the true dependence-distance curve with small bias and acceptable variability, then the empirical winter–summer comparison can be interpreted with greater confidence. Second, the simulation clarifies the direction and magnitude of finite-sample uncertainty. In particular, it shows whether estimated seasonal differences are likely to be detectable for realistic sample sizes, or whether only large differences in dependence range can be recovered reliably.

The simulation therefore provides a bridge between the theoretical estimands and the empirical application. The empirical analysis observes only one finite sample of Dutch daily maximum wind gusts. The simulation asks whether, under comparable finite-sample conditions, the proposed methods can recover a known dependence structure. This validation step is essential for interpreting the empirical dependence-distance curves as evidence about seasonal differences in spatial extremal dependence.

A natural extension, left for future work, is to relax the independence of the simulated daily fields by introducing short-range temporal dependence between consecutive days, for example through a max-autoregressive or storm-clustered construction. Such a design would let the simulation test the year or seasonal-block bootstrap of Section 3.10 under the same multi-day storm dependence present in the KNMI data, rather than only under independent replications. This is not required to answer the main research question, which concerns whether the pairwise estimators can recover a known spatial dependence difference, but it would further strengthen the link between the simulated and empirical uncertainty quantification.

A Mathematical details for pairwise extremal dependence estimators

A.1 Probability integral transform

We prove that the marginal transform $U_{i,t} = F_i(X_{i,t})$ has a standard uniform distribution when F_i is continuous.

Proof. Let $X_{i,t}$ be a real-valued random variable with continuous distribution function F_i . Define

$$U_{i,t} = F_i(X_{i,t}).$$

For $u < 0$,

$$\Pr(U_{i,t} \leq u) = 0.$$

For $u > 1$,

$$\Pr(U_{i,t} \leq u) = 1.$$

Let $u \in [0, 1]$. Define the generalized inverse

$$F_i^{-1}(u) = \inf\{x \in \mathbb{R} : F_i(x) \geq u\}.$$

Since F_i is continuous,

$$F_i(F_i^{-1}(u)) = u.$$

Then

$$\begin{aligned} \Pr(U_{i,t} \leq u) &= \Pr(F_i(X_{i,t}) \leq u) \\ &= \Pr(X_{i,t} \leq F_i^{-1}(u)) \\ &= F_i(F_i^{-1}(u)) \\ &= u. \end{aligned}$$

Hence

$$\Pr(U_{i,t} \leq u) = u, \quad u \in [0, 1].$$

Therefore

$$U_{i,t} \sim \text{Uniform}(0, 1).$$

For two stations i and j , define

$$U_{i,t} = F_i(X_{i,t}), \quad U_{j,t} = F_j(X_{j,t}).$$

Then

$$U_{i,t} \sim \text{Uniform}(0, 1), \quad U_{j,t} \sim \text{Uniform}(0, 1).$$

The marginal distributions are therefore equal after transformation, and the joint distribution of $(U_{i,t}, U_{j,t})$ contains only dependence information. $\square$

A.2 Relation between χ_{ij} and θ_{ij} under max-stability

We prove that, under bivariate max-stability with standard uniform margins, the upper tail dependence coefficient and the extremal coefficient satisfy $\chi_{ij} = 2 - \theta_{ij}$.

Proof. Let (U_i, U_j) have standard uniform margins. Assume the bivariate extreme-value copula representation

$$\Pr(U_i \leq u, U_j \leq u) = u^{\theta_{ij}}, \quad u \in (0, 1),$$

where

$$\theta_{ij} \in [1, 2].$$

The upper tail dependence coefficient is

$$\chi_{ij} = \lim_{u \uparrow 1} \Pr(U_j > u \mid U_i > u).$$

By the definition of conditional probability,

$$\Pr(U_j > u \mid U_i > u) = \frac{\Pr(U_i > u, U_j > u)}{\Pr(U_i > u)}.$$

Since $U_i \sim \text{Uniform}(0, 1)$,

$$\Pr(U_i > u) = 1 - u.$$

By inclusion-exclusion,

$$\begin{aligned} \Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \leq u) - \Pr(U_j \leq u) + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - u - u + u^{\theta_{ij}} \\ &= 1 - 2u + u^{\theta_{ij}}. \end{aligned}$$

Therefore

$$\Pr(U_j > u \mid U_i > u) = \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Thus

$$\chi_{ij} = \lim_{u \uparrow 1} \frac{1 - 2u + u^{\theta_{ij}}}{1 - u}.$$

Both numerator and denominator converge to zero:

$$\lim_{u \uparrow 1} (1 - 2u + u^{\theta_{ij}}) = 1 - 2 + 1 = 0,$$

$$\lim_{u \uparrow 1} (1 - u) = 0.$$

By l'Hopital's rule,

$$\begin{aligned} \chi_{ij} &= \lim_{u \uparrow 1} \frac{-2 + \theta_{ij} u^{\theta_{ij}-1}}{-1} \\ &= \frac{-2 + \theta_{ij}}{-1} \\ &= 2 - \theta_{ij}. \end{aligned}$$

Hence

$$\boxed{\chi_{ij} = 2 - \theta_{ij}.$$

The endpoints are consistent with the interpretation:

$$\theta_{ij} = 1 \implies \chi_{ij} = 1,$$

$$\theta_{ij} = 2 \implies \chi_{ij} = 0.$$

□

A.3 Extremal coefficient as the exponent measure evaluated at $(1, 1)$

We prove that, for a bivariate max-stable distribution with unit Fréchet margins, the pairwise extremal coefficient is $\theta_{ij} = V_{ij}(1, 1)$.

Proof. Let (Z_i, Z_j) denote the bivariate vector

$$(Z_i, Z_j) = (Z(s_i), Z(s_j)).$$

Assume unit Fréchet margins:

$$\Pr(Z_i \leq z) = \exp(-1/z), \quad z > 0,$$

$$\Pr(Z_j \leq z) = \exp(-1/z), \quad z > 0.$$

Let the joint distribution have exponent measure representation

$$\Pr(Z_i \leq z_i, Z_j \leq z_j) = \exp\{-V_{ij}(z_i, z_j)\}, \quad z_i, z_j > 0.$$

Set

$$z_i = z, \quad z_j = z.$$

Then

$$\Pr(Z_i \leq z, Z_j \leq z) = \exp\{-V_{ij}(z, z)\}.$$

For a max-stable distribution with unit Fréchet margins, the exponent measure is homogeneous of order -1 :

$$V_{ij}(az_i, az_j) = a^{-1}V_{ij}(z_i, z_j), \quad a > 0.$$

Take

$$a = z, \quad z_i = 1, \quad z_j = 1.$$

Then

$$\begin{aligned} V_{ij}(z, z) &= V_{ij}(z \cdot 1, z \cdot 1) \\ &= z^{-1}V_{ij}(1, 1). \end{aligned}$$

Therefore

$$\begin{aligned} \Pr(Z_i \leq z, Z_j \leq z) &= \exp\{-z^{-1}V_{ij}(1, 1)\} \\ &= \exp\left\{-\frac{V_{ij}(1, 1)}{z}\right\}. \end{aligned}$$

The pairwise extremal coefficient θ_{ij} is defined by

$$\Pr(Z_i \leq z, Z_j \leq z) = \Pr(Z_i \leq z)^{\theta_{ij}}.$$

Since

$$\Pr(Z_i \leq z) = \exp(-1/z),$$

we have

$$\begin{aligned} \Pr(Z_i \leq z)^{\theta_{ij}} &= [\exp(-1/z)]^{\theta_{ij}} \\ &= \exp(-\theta_{ij}/z). \end{aligned}$$

Equating the two expressions for the same joint probability gives

$$\exp\left\{-\frac{V_{ij}(1, 1)}{z}\right\} = \exp\left\{-\frac{\theta_{ij}}{z}\right\}.$$

Taking logarithms gives

$$-\frac{V_{ij}(1, 1)}{z} = -\frac{\theta_{ij}}{z}.$$

Multiplying by $-z$ gives

$$V_{ij}(1, 1) = \theta_{ij}.$$

Hence

$$\boxed{\theta_{ij} = V_{ij}(1, 1)}.$$

The bounds follow from the two limiting cases. Perfect dependence gives

$$Z_i = Z_j,$$

so

$$\Pr(Z_i \leq z, Z_j \leq z) = \Pr(Z_i \leq z) = \exp(-1/z),$$

which implies

$$\theta_{ij} = 1.$$

Independence gives

$$\begin{aligned}\Pr(Z_i \leq z, Z_j \leq z) &= \Pr(Z_i \leq z) \Pr(Z_j \leq z) \\ &= \exp(-1/z) \exp(-1/z) \\ &= \exp(-2/z),\end{aligned}$$

which implies

$$\theta_{ij} = 2.$$

Therefore

$$1 \leq \theta_{ij} \leq 2.$$

□

A.4 F-madogram relation to the extremal coefficient

We prove that, under bivariate max-stability with standard uniform margins, the F-madogram satisfies $\theta_{ij} = (1 + 2\nu_F)/(1 - 2\nu_F)$.

Proof. Let

$$U_i = F_i(Z_i), \quad U_j = F_j(Z_j),$$

where U_i and U_j have standard uniform margins. The F-madogram is

$$\nu_F(i, j) = \frac{1}{2} \mathbb{E}|U_i - U_j|.$$

Use the identity

$$|a - b| = a + b - 2 \min(a, b), \quad a, b \in [0, 1].$$

Then

$$\begin{aligned}\mathbb{E}|U_i - U_j| &= \mathbb{E}(U_i + U_j - 2 \min(U_i, U_j)) \\ &= \mathbb{E}U_i + \mathbb{E}U_j - 2\mathbb{E} \min(U_i, U_j).\end{aligned}$$

Since $U_i, U_j \sim \text{Uniform}(0, 1)$,

$$\mathbb{E}U_i = \frac{1}{2}, \quad \mathbb{E}U_j = \frac{1}{2}.$$

Therefore

$$\mathbb{E}|U_i - U_j| = 1 - 2\mathbb{E} \min(U_i, U_j).$$

Thus

$$\nu_F(i, j) = \frac{1}{2} - \mathbb{E} \min(U_i, U_j).$$

It remains to compute $\mathbb{E} \min(U_i, U_j)$. For any random variable $Y \in [0, 1]$,

$$\mathbb{E}Y = \int_0^1 \Pr(Y > u) du.$$

Set

$$Y = \min(U_i, U_j).$$

Then

$$\begin{aligned}\mathbb{E} \min(U_i, U_j) &= \int_0^1 \Pr(\min(U_i, U_j) > u) du \\ &= \int_0^1 \Pr(U_i > u, U_j > u) du.\end{aligned}$$

By inclusion-exclusion,

$$\begin{aligned}\Pr(U_i > u, U_j > u) &= 1 - \Pr(U_i \leq u) - \Pr(U_j \leq u) + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - u - u + \Pr(U_i \leq u, U_j \leq u) \\ &= 1 - 2u + \Pr(U_i \leq u, U_j \leq u).\end{aligned}$$

Under bivariate max-stability with extremal coefficient θ_{ij} ,

$$\Pr(U_i \leq u, U_j \leq u) = u^{\theta_{ij}}.$$

Therefore

$$\begin{aligned}\mathbb{E} \min(U_i, U_j) &= \int_0^1 \{1 - 2u + u^{\theta_{ij}}\} du \\ &= \int_0^1 1 du - 2 \int_0^1 u du + \int_0^1 u^{\theta_{ij}} du \\ &= 1 - 2 \left[\frac{u^2}{2} \right]_0^1 + \left[\frac{u^{\theta_{ij}+1}}{\theta_{ij}+1} \right]_0^1 \\ &= 1 - 1 + \frac{1}{\theta_{ij}+1} \\ &= \frac{1}{\theta_{ij}+1}.\end{aligned}$$

Hence

$$\begin{aligned}\nu_F(i, j) &= \frac{1}{2} - \frac{1}{\theta_{ij}+1} \\ &= \frac{\theta_{ij}+1}{2(\theta_{ij}+1)} - \frac{2}{2(\theta_{ij}+1)} \\ &= \frac{\theta_{ij}-1}{2(\theta_{ij}+1)}.\end{aligned}$$

Thus

$$2\nu_F(i, j) = \frac{\theta_{ij}-1}{\theta_{ij}+1}.$$

Solve for θ_{ij} :

$$\begin{aligned}2\nu_F(\theta_{ij}+1) &= \theta_{ij}-1, \\ 2\nu_F\theta_{ij} + 2\nu_F &= \theta_{ij}-1, \\ \theta_{ij} - 2\nu_F\theta_{ij} &= 1 + 2\nu_F, \\ \theta_{ij}(1 - 2\nu_F) &= 1 + 2\nu_F, \\ \theta_{ij} &= \frac{1 + 2\nu_F}{1 - 2\nu_F}.\end{aligned}$$

Hence

$$\boxed{\theta_{ij} = \frac{1 + 2\nu_F(i, j)}{1 - 2\nu_F(i, j)}}.$$

Replacing $\nu_F(i, j)$ by the empirical F-madogram

$$\hat{\nu}_F(i, j; c) = \frac{1}{2B^{(c)}} \sum_{b=1}^{B^{(c)}} \left| \hat{F}_i^{(c)}(M_{i,b}^{(c)}) - \hat{F}_j^{(c)}(M_{j,b}^{(c)}) \right|$$

gives the estimator

$$\hat{\theta}_{ij}^{(c)} = \frac{1 + 2\hat{\nu}_F(i, j; c)}{1 - 2\hat{\nu}_F(i, j; c)}.$$

□

A.5 Consistency of the empirical finite-level tail-dependence estimator

We prove that, under within-season stationarity and ergodicity, the empirical estimator $\widehat{\chi}_{ij}^{(c)}(u)$ converges to the finite-level conditional tail probability $\Pr(U_j^{(c)} > u \mid U_i^{(c)} > u)$.

Proof. Fix a season

$$c \in \{W, S\}.$$

Fix a pair of stations

$$(i, j).$$

Let

$$\{(U_{i,t}^{(c)}, U_{j,t}^{(c)}) : t \in T_c\}$$

be the season- c transformed pair process. Assume this process is strictly stationary and ergodic. Fix

$$u \in (0, 1).$$

Define

$$A_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\},$$

and

$$B_t^{(c)}(u) = \mathbf{1}\{U_{i,t}^{(c)} > u\}.$$

Since $(U_{i,t}^{(c)}, U_{j,t}^{(c)})$ is strictly stationary and ergodic, any measurable transformation of it is also strictly stationary and ergodic. Therefore

$$\{A_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic, and

$$\{B_t^{(c)}(u) : t \in T_c\}$$

is strictly stationary and ergodic. Moreover,

$$0 \leq A_t^{(c)}(u) \leq 1, \quad 0 \leq B_t^{(c)}(u) \leq 1.$$

Hence

$$\mathbb{E}|A_t^{(c)}(u)| < \infty, \quad \mathbb{E}|B_t^{(c)}(u)| < \infty.$$

By the ergodic theorem,

$$\frac{1}{|T_c|} \sum_{t \in T_c} A_t^{(c)}(u) \xrightarrow{p} \mathbb{E}A_t^{(c)}(u),$$

and

$$\frac{1}{|T_c|} \sum_{t \in T_c} B_t^{(c)}(u) \xrightarrow{p} \mathbb{E}B_t^{(c)}(u).$$

By the definition of $A_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}A_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u). \end{aligned}$$

By the definition of $B_t^{(c)}(u)$,

$$\begin{aligned} \mathbb{E}B_t^{(c)}(u) &= \mathbb{E}\mathbf{1}\{U_{i,t}^{(c)} > u\} \\ &= \Pr(U_{i,t}^{(c)} > u). \end{aligned}$$

Since $U_{i,t}^{(c)}$ has a standard uniform margin,

$$\Pr(U_{i,t}^{(c)} > u) = 1 - u.$$

For $u \in (0, 1)$,

$$1 - u > 0.$$

The empirical finite-level tail-dependence estimator is

$$\hat{\chi}_{ij}^{(c)}(u) = \frac{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u\}}{\sum_{t \in T_c} \mathbf{1}\{U_{i,t}^{(c)} > u\}}.$$

Using $A_t^{(c)}(u)$ and $B_t^{(c)}(u)$,

$$\begin{aligned} \hat{\chi}_{ij}^{(c)}(u) &= \frac{\sum_{t \in T_c} A_t^{(c)}(u)}{\sum_{t \in T_c} B_t^{(c)}(u)} \\ &= \frac{|T_c|^{-1} \sum_{t \in T_c} A_t^{(c)}(u)}{|T_c|^{-1} \sum_{t \in T_c} B_t^{(c)}(u)}. \end{aligned}$$

By the continuous mapping theorem,

$$\begin{aligned} \hat{\chi}_{ij}^{(c)}(u) &\xrightarrow{p} \frac{\Pr(U_{i,t}^{(c)} > u, U_{j,t}^{(c)} > u)}{\Pr(U_{i,t}^{(c)} > u)} \\ &= \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u). \end{aligned}$$

Therefore

$$\boxed{\hat{\chi}_{ij}^{(c)}(u) \xrightarrow{p} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)}$$

for every fixed $u \in (0, 1)$. If the limit

$$\chi_{ij}^{(c)} = \lim_{u \uparrow 1} \Pr(U_{j,t}^{(c)} > u \mid U_{i,t}^{(c)} > u)$$

exists, then the finite-level probability estimated above is the finite- u approximation to the asymptotic tail-dependence coefficient. $\square$

References

- Balkema, A. A. and L. de Haan (1974). “Residual Life Time at Great Age”. In: *The Annals of Probability* 2.5, pp. 792–804. DOI: 10.1214/aop/1176996548.
- Brown, Bruce M. and Sidney I. Resnick (1977). “Extreme values of independent stochastic processes”. In: *Journal of Applied Probability* 14.4, pp. 732–739. DOI: 10.2307/3213322.
- Coles, Stuart, Janet Heffernan, and Jonathan Tawn (1999). “Dependence measures for extreme value analyses”. In: *Extremes* 2.4, pp. 339–365. DOI: 10.1023/A:1009963131610.
- Cooley, Daniel, Jessi Cisewski, Robert J. Erhardt, Soyoung Jeon, Elizabeth Mannshardt, Bernard Oguna Omolo, and Ying Sun (2012). “A Survey of Spatial Extremes: Measuring Spatial Dependence and Modeling Spatial Effects”. In: *REVSTAT – Statistical Journal* 10.1, pp. 135–165. DOI: 10.57805/revstat.v10i1.114.
- Cooley, Daniel, Philippe Naveau, and Philippe Poncet (2006). “Variograms for Spatial Max-Stable Random Fields”. In: *Dependence in Probability and Statistics*. Ed. by Patrice Bertail, Paul Doukhan, and Philippe Soulier. Vol. 187. Lecture Notes in Statistics. New York: Springer, pp. 373–390. DOI: 10.1007/0-387-36062-X_17.
- Davison, A. C., S. A. Padoan, and M. Ribatet (2012). “Statistical Modeling of Spatial Extremes”. In: *Statistical Science* 27.2, pp. 161–186. DOI: 10.1214/11-STS376.
- Deltacommissie (1960). *Rapport Deltacommissie: Eindverslag en interimadviezen (Report of the Delta Committee: Final report and interim advices)*. The Hague, The Netherlands: Staatsdrukkerij- en Uitgeverijbedrijf.
- Dombry, Clément, Sebastian Engelke, and Marco Oesting (2016). “Exact simulation of max-stable processes”. In: *Biometrika* 103.2, pp. 303–317. DOI: 10.1093/biomet/asw008.
- Fawcett, Lee and David Walshaw (2006). “Markov chain models for spatial extremes”. In: *Environmetrics* 17.2, pp. 199–216. DOI: 10.1002/env.754.
- Ferro, Christopher A. T. and Johan Segers (2003). “Inference for clusters of extreme values”. In: *Journal of the Royal Statistical Society: Series B (Statistical Methodology)* 65.2, pp. 545–556. DOI: 10.1111/1467-9868.00401.
- Fisher, R. A. and L. H. C. Tippett (1928). “Limiting Forms of the Frequency Distribution of the Largest or Smallest Member of a Sample”. In: *Mathematical Proceedings of the Cambridge Philosophical Society* 24.2, pp. 180–190. DOI: 10.1017/S0305004100015681.
- Fonseca Cerda, M. d. S., H. de Moel, D. van Ederen, J. C. J. H. Aerts, W. J. W. Botzen, and T. Haer (2025). “Empirical Evaluation of Windstorm Losses and Meteorological Variables over the Netherlands”. In: *Natural Hazards* 121.5, pp. 6085–6106. DOI: 10.1007/s11069-024-07024-y.
- Gerritsen, Herman (2005). “What happened in 1953? The Big Flood in the Netherlands in retrospect”. In: *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences* 363.1831, pp. 1271–1291. DOI: 10.1098/rsta.2005.1568.
- Gliksmann, D., P. Averbeck, N. Becker, B. Gardiner, V. Goldberg, J. Grieger, et al. (2023). “A European perspective on wind and storm damage: From the meteorological background to index-based approaches to assess impacts”. In: *Natural Hazards and Earth System Sciences* 23, pp. 2171–2201.
- Gnedenko, B. V. (1943). “Sur la distribution limite du terme maximum d’une série aléatoire”. In: *Annals of Mathematics* 44.3, pp. 423–453. DOI: 10.2307/1968974.
- Haan, Laurens de and Teresa T. Pereira (2006). “Spatial Extremes: Models for the Stationary Case”. In: *The Annals of Statistics* 34.1, pp. 146–168. DOI: 10.1214/009053605000000886.
- Huser, Raphaël and Marc G. Genton (2016). “Non-Stationary Dependence Structures for Spatial Extremes”. In: *Journal of Agricultural, Biological, and Environmental Statistics* 21, pp. 470–491. DOI: 10.1007/s13253-016-0247-4.
- Huser, Raphaël and Jennifer L. Wadsworth (2022). “Advances in Statistical Modeling of Spatial Extremes”. In: *WIREs Computational Statistics* 14.1, e1537. DOI: 10.1002/wics.1537.
- Little, Alexander S., Matthew D. K. Priestley, and Jennifer L. Catto (2023). “Future Increased Risk from Extratropical Windstorms in Northern Europe”. In: *Nature Communications* 14, p. 4434. DOI: 10.1038/s41467-023-40102-6.
- Mohr, Susanna, Michael Kunz, Alexandra Richter, and Bodo Ruck (2017). “Statistical Characteristics of Convective Wind Gusts in Germany”. In: *Natural Hazards and Earth System Sciences* 17, pp. 957–969. DOI: 10.5194/nhess-17-957-2017.
- Padoan, S. A., M. Ribatet, and S. A. Sisson (2010). “Likelihood-Based Inference for Max-Stable Processes”. In: *Journal of the American Statistical Association* 105.489, pp. 263–277. DOI: 10.1198/jasa.2009.tm08577.

- Pickands, James (1975). “Statistical Inference Using Extreme Order Statistics”. In: *The Annals of Statistics* 3.1, pp. 119–131. DOI: 10.1214/aos/1176343003.
- Sharkey, Paul, Jonathan A. Tawn, and Simon J. Brown (2020). “Modelling the Spatial Extent and Severity of Extreme European Windstorms”. In: *Journal of the Royal Statistical Society: Series C (Applied Statistics)* 69.2, pp. 223–250. DOI: 10.1111/rssc.12391.
- Verbond van Verzekeraars (2017). *Hoofd boven water: Verzekeren van schade in een veranderend klimaat*. Tech. rep. Geraadpleegd op 30 mei 2026. Den Haag: Verbond van Verzekeraars. URL: <https://www.verzekeraars.nl/media/3643/hoofd-boven-water-verzekeren-van-schade-in-een-veranderend-klimaat.pdf>.
- Wadsworth, Jennifer L. and Jonathan A. Tawn (2012). “Dependence Modelling for Spatial Extremes”. In: *Biometrika* 99.2, pp. 253–272. DOI: 10.1093/biomet/asr080.

Table 1: All available variables in hourly-in-situ-meteorological-observations.

Parameter	Description	Unit	Observed property	Comment
DD	Past 10 minute mean wind direction, representative for 10 metres, in degrees; 360 = north; 90 = east; 180 = south; 270 = west; 0 = calm; 990 = variable	°	Wind from direction	
DR	Hourly precipitation duration, in hours	h	Duration of precipitation	
EE	Hourly mean vapour pressure, in hectopascal	hPa	Water vapour partial pressure in air	
FF	Past 10 minute mean wind speed, representative for 10 metres, in metres per second	m/s	Wind speed	
FH	Hourly mean wind speed, representative for 10 metres, in metres per second	m/s	Wind speed	
FX	Hourly maximum wind gust, representative for 10 metres, in metres per second	m/s	Wind speed of gust	
IX	Hourly indicator present weather code; 1 = manned and recorded using code from visual observations; 2; 3 = manned and omitted, no significant weather phenomenon to report or not available; 4 = automatically recorded using code from visual observations; 5; 6 = automatically omitted, no significant weather phenomenon to report or not available; 7 = automatically set using code from automated observations	–	Indicator present weather code	
N	Past 10 minute cloud cover, in okta; 9 = sky invisible	okta	Cloud area fraction	
P	Past 1 minute mean air pressure at sea level	hPa	Air pressure at mean sea level	
Q	Hourly global solar radiation, in joules per square centimetre	J/cm ²	Integral with respect to time of surface downwelling short-wave flux in air	
RH	Hourly precipitation amount, in millimetres	mm	Precipitation amount	¹
SQ	Hourly sunshine duration, calculated from global solar radiation, in hours	h	Duration of sunshine	²
T	Past 1 minute air temperature at 1.50 metres, in degrees Celsius	°C	Air temperature	
T10N	Past 6 hour minimum air temperature at 10 centimetres, in degrees Celsius	°C	Air temperature	
TD	Past 1 minute dew point temperature at 1.50 metres, in degrees Celsius	°C	Dew point temperature	
U	Past 1 minute relative atmospheric humidity at the time of observation, at 1.50 metres, as percentage	%	Relative humidity	
VV	Past 10 minute horizontal visibility; 0 = less than 100 m; 1 = 100–200 m; 2 = 200–300 m; ...; 49 = 4900–5000 m; 50 = 5–6 km; 56 = 6–7 km; 57 = 7–8 km; ...; 79 = 29–30 km; 80 = 30–35 km; 81 = 35–40 km; ...; 89 = more than 70 km	–	Visibility in air	
W1	Hourly indicator of fog; 0 = no occurrence; 1 = occurred	–	Fog	
W2	Hourly indicator of rainfall; 0 = no occurrence; 1 = occurred	–	Rain	
W3	Hourly indicator of snow; 0 = no occurrence; 1 = occurred	–	Snow	
W5	Hourly indicator of thunder; 0 = no occurrence; 1 = occurred	–	Thunder	

Continued on next page

Table 1: All available variables in `hourly-in-situ-meteorological-observations` – continued.

Parameter	Description	Unit	Observed property	Comment
W6	Hourly indicator of ice formation; 0 = no occurrence; 1 = occurred	–	Ice formation	
WW	Hourly present weather code, aggregated with an algorithm using 10-minute <code>ww</code> WaWa values from the preceding 6 hours; WMO table 4680	–	Present weather code	

Notes:

¹ For precipitation amounts the detection minimum is 0.05 millimetres.

² For sunshine duration the measured minimum is 0.05 hours.